

Chapter 11 - Grid

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→ What is CSS Grid?

- CSS Grid is a layout system used to arrange elements in rows and columns.

X Grid ka use element ko rows aur columns mein arrange karne ke liye hota hai.

column			
↓	↓	↓	
1	2	3	← row
4	5	6	← row
7	8	9	← row

(i) `display: grid`

- Grid start karne ke liye.

```
.container {  
  display: grid  
}
```

```
<div class="container">  
  <div> 1 </div>  
  <div> 2 </div>  
</div>
```

`display: grid` (parent ko Grid Container bana deta hai).

ii) Grid Container & Grid Items.

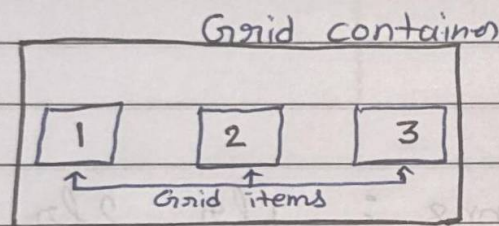
- A grid container is the parent element where `display: grid` is applied, while grid items are the child elements placed inside the grid container.

example:

```
<div class="container">
  <div> 1 </div>
  <div> 2 </div>
  <div> 3 </div>
</div>
```

CSS

```
.container {
  display: grid;
}
```



- Parent → Grid Container
- Direct children → Grid Items

iii) `grid-template-columns`

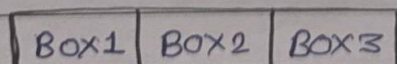
- This property defines how many columns we want

✗ Ye decide karta hai ki Grid me kitna columns honge
~~Example~~ aur unka size kya hoga.

Example:

```
.container {
  display: grid;
  grid-template-columns: 200px 200px 200px;
}
```

Output:



iv) fr Unit

- fr means a fraction of the available space.
- * fr available space ko parts me divide karita hai.

example:

grid-template-columns: 1fr 1fr 1fr;

Output:

1fr	1fr	1fr
-----	-----	-----

* 1fr 1fr 1fr
= 3 equal columns

example:

grid-template-columns: 1fr 2fr 1fr;

Output:

1fr	2fr	1fr
-----	-----	-----

* Total part = 4
• First = 1 part
• Second = 2 part
• Third = 1 part

v) grid-template-rows.

- Defines the size of grid rows.

X Grid ki rows ka size decide karta hai.

Example:

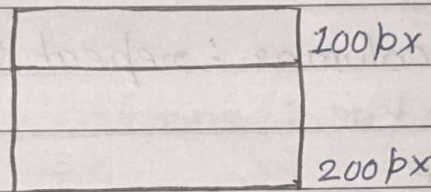
```
container {
```

```
display: grid;
```

```
grid-template-rows: 100px 200px;
```

```
}
```

Output:



vi) gap

- Adds space between grid items.

X Grid items ke beech me spacing deta hai

example:

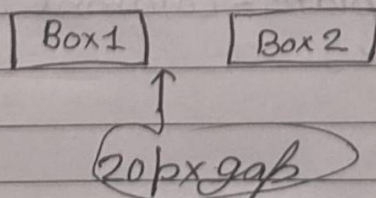
```
container {
```

```
display: grid;
```

```
gap: 20px;
```

```
}
```

Output:



<vii> repeat()

- Repeats columns or rows automatically.
- X Same value ko baar-baar likhne ki jagah repeat karta hai

Before (Pehle aise likhte the)

```
grid-template-columns: 1fr 1fr 1fr;
```

Now (Ab aise likh sakte hai)

```
grid-template-columns: repeat(3, 1fr);
```

➔ Real Example

```
<div class="container">
  <div class="box"> Box 1 </div>
  <div class="box"> Box 2 </div>
  <div class="box"> Box 3 </div>
  <div class="box"> Box 4 </div>
  <div class="box"> Box 5 </div>
  <div class="box"> Box 6 </div>
</div>
```

Css

```
.container {
```

```
  display: grid;
```

```
  grid-template-columns: repeat(3, 1fr);
```

```
  gap: 20px;
```

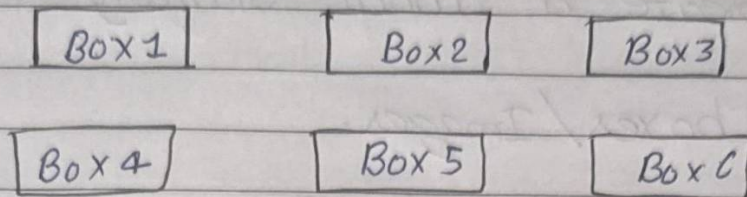
```
}
```

```
.box {
```

```
  border: 2px solid black;
```

```
  padding: 20px;
```

Output :



→ Cheat Sheet

```
/* Create Grid */
```

```
display : grid;
```

```
/* Equal Columns */
```

```
grid-template-columns : repeat(3, 1fr);
```

```
/* Rows */
```

```
grid-template-rows : 100px 200px;
```

```
/* Space Between items */
```

```
gap : 20px;
```

```
/* 3 Columns Manually */
```

```
grid-template-columns : 1fr 1fr 1fr
```

```
/* Different Column Size */
```

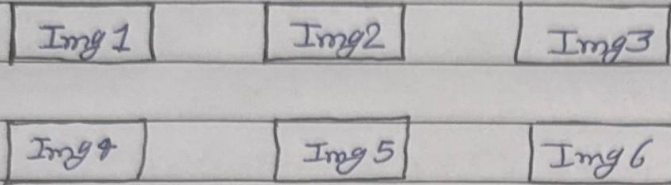
```
grid-template-columns : 33% 1fr 2fr 1fr;
```



Practice.

Task: Create a Photo Gallery

Create 6 boxes/Images.



CSS →

```
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 20px;
```